Year 10 Foundation

REVISION PACK

Progress Check 3

ANSWERS

Units in this pack

- Number 3
- Algebra 2
- Graphs 2

Teacher copy: worked answers to every practice question in the revision pack. Working is set out one line under another, as students are expected to show it.

Number 3 · Laws of indices (Lesson 6.1)

Q	Working	Answer
writing as a single power		
1.1	**(a)** 4^{5+2} **(b)** 8^{10-3} **(c)** $5^{2\times 3}$	**(a)** 4^{7} **(b)** 8^{7} **(c)** 5^{6}
1.2	**(a)** 6^{2+7} **(b)** 9^{8-5} **(c)** $7^{4\times 2}$	**(a)** 6^{9} **(b)** 9^{3} **(c)** 7^{8}
1.3	**(a)** 3^{4+6} **(b)** 2^{9-2} **(c)** $2^{5\times 3}$	**(a)** 3^{10} **(b)** 2^{7} **(c)** 2^{15}
1.4	**(a)** 10^{3+5} **(b)** 5^{12-4} **(c)** $4^{3\times 4}$	**(a)** 10^{8} **(b)** 5^{8} **(c)** 4^{12}
1.5	**(a)** 7^{2+8} **(b)** 3^{11-6} **(c)** $6^{2\times 6}$	**(a)** 7^{10} **(b)** 3^{5} **(c)** 6^{12}
1.6	**(a)** 2^{6+5} **(b)** 6^{9-3} **(c)** $9^{3\times 3}$	**(a)** 2^{11} **(b)** 6^{6} **(c)** 9^{9}
negative and zero powers		
1.7	**(a)** $4^{-2} = \frac{1}{4^{2}}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{16}$ **(b)** **1**

1.8	**(a)** $2^{-4} = \frac{1}{2^4}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{16}$ **(b) 1**
1.9	**(a)** $10^{-2} = \frac{1}{10^2}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{100}$ **(b) 1**
1.10	**(a)** $3^{-3} = \frac{1}{3^3}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{27}$ **(b) 1**
1.11	**(a)** $5^{-2} = \frac{1}{5^2}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{25}$ **(b) 1**
1.12	**(a)** $6^{-1} = \frac{1}{6^1}$ **(b)** any number to the power 0 is 1	**(a)** $\frac{1}{6}$ **(b) 1**

Algebra 2 · Indices in algebra (Lesson 7.1)

Q	Working	Answer
simplifying with index laws		
2.1	**(a)** x^{2+7} **(b)** y^{10-4} **(c)** $5^2 = 25,\ 2 \times 2 = 4$ **(d)** $8 \div 2 = 4,\ m^{7-3},\ n^{3-1}$	**(a)** x^9 **(b)** y^6 **(c)** $25x^4$ **(d)** $4m^4n^2$
2.2	**(a)** a^{5+3} **(b)** b^{7-2} **(c)** $2^3 = 8,\ 4 \times 3 = 12$ **(d)** $15 \div 3 = 5,\ x^{5-2},\ y^{4-1}$	**(a)** a^8 **(b)** b^5 **(c)** $8a^{12}$ **(d)** $5x^3y^3$

2.3	**(a)** p^{4+6} **(b)** q^{9-3} **(c)** $4^2 = 16,\ 5 \times 2 = 10$ **(d)** $12 \div 4 = 3,\ p^{9-4},\ q^{2-2}$	**(a)** p^{10} **(b)** q^{6} **(c)** $16y^{10}$ **(d)** $3p^{5}$
2.4	**(a)** m^{3+8} **(b)** n^{12-5} **(c)** $3^3 = 27,\ 2 \times 3 = 6$ **(d)** $20 \div 5 = 4,\ a^{6-2},\ b^{4-3}$	**(a)** m^{11} **(b)** n^{7} **(c)** $27x^{6}$ **(d)** $4a^{4}b$
2.5	**(a)** x^{6+4} **(b)** y^{11-2} **(c)** $6^2 = 36,\ 3 \times 2 = 6$ **(d)** $18 \div 9 = 2,\ m^{5-4},\ n^{3-1}$	**(a)** x^{10} **(b)** y^{9} **(c)** $36b^{6}$ **(d)** $2mn^{2}$
2.6	**(a)** t^{7+1} **(b)** k^{15-5} **(c)** $2^4 = 16,\ 5 \times 4 = 20$ **(d)** $14 \div 7 = 2,\ x^{8-3},\ y^{5-2}$	**(a)** t^{8} **(b)** k^{10} **(c)** $16c^{20}$ **(d)** $2x^{5}y^{3}$

Graphs 2 · Conversion graphs (Lesson 8.1)

Q	Working	Answer
using a conversion graph		
3.1	30 KD = €90 (from the graph) €90 × 10 = €900	**(a) €60** **(b) 30 KD** **(c) €900**
3.2	40 KD = £100 (from the graph) £100 × 10 = £1000	**(a) £40** **(b) 28 KD** **(c) £1000**
3.3	20 KD = AED 240 (from the graph) AED 240 × 10 = AED 2400	**(a) AED 180** **(b) 8 KD** **(c) AED 2400**
3.4	15 KD = SAR 60 (from the graph) SAR 60 × 10 = SAR 600	**(a) SAR 100** **(b) 15 KD** **(c) SAR 600**

Graphs 2 · Distance-time graphs (Lesson 8.2)

Q	Working	Answer
reading a distance-time graph		
4.1	$35 - 20 = 15$ 8 km in 20 min $8 \div \frac{1}{3} = 24$	**(a) 15 minutes** **(b) 4 km** **(c) 24 km/h**
4.2	$60 - 40 = 20$ 10 km in 30 min $10 \div \frac{1}{2} = 20$	**(a) 20 minutes** **(b) 5 km** **(c) 20 km/h**
4.3	$45 - 15 = 30$ 6 km in 15 min $6 \div \frac{1}{4} = 24$	**(a) 30 minutes** **(b) 2 km** **(c) 24 km/h**
4.4	$50 - 30 = 20$ 9 km in 30 min $9 \div \frac{1}{2} = 18$	**(a) 20 minutes** **(b) 3 km** **(c) 18 km/h**

Graphs 2 · Speed-time graphs (Lesson 8.3)

Q	Working	Answer
acceleration and constant speed		
5.1	$(a)\ 10 \div 5 = 2$ $(b)\ (15 - 5) \times 10 = 100$	**(a) 2 m/s²** **(b) 100 m**
5.2	$(a)\ 12 \div 4 = 3$ $(b)\ (10 - 4) \times 12 = 72$	**(a) 3 m/s²** **(b) 72 m**
5.3	$(a)\ 18 \div 6 = 3$ $(b)\ (16 - 6) \times 18 = 180$	**(a) 3 m/s²** **(b) 180 m**
5.4	$(a)\ 20 \div 10 = 2$ $(b)\ (20 - 10) \times 20 = 200$	**(a) 2 m/s²** **(b) 200 m**

Graphs 2 · Mid-points and perpendicular lines (Lessons 8.4, 8.5)

Q	Working	Answer
mid-point of a line		
6.1	$\left(\frac{4+10}{2}, \frac{1+7}{2}\right)$	**(7, 4)**
6.2	$\left(\frac{1+9}{2}, \frac{6+12}{2}\right)$	**(5, 9)**
6.3	$\left(\frac{3+9}{2}, \frac{2+12}{2}\right)$	**(6, 7)**
6.4	$\left(\frac{2+6}{2}, \frac{4+10}{2}\right)$	**(4, 7)**
6.5	$\left(\frac{0+8}{2}, \frac{3+5}{2}\right)$	**(4, 4)**
6.6	$\left(\frac{5+11}{2}, \frac{1+9}{2}\right)$	**(8, 5)**
perpendicular lines		
6.7	**(a)** $-1 \div 2 = -\frac{1}{2}$ **(b)** $m_1 = 4,\ m_2 = -\frac{1}{4}$ $4 \times (-\frac{1}{4}) = -1$	**(a)** $-\frac{1}{2}$ **(b) gradients multiply to −1, so perpendicular**
6.8	**(a)** $-1 \div 6 = -\frac{1}{6}$ **(b)** $m_1 = 5,\ m_2 = -\frac{1}{5}$ $5 \times (-\frac{1}{5}) = -1$	**(a)** $-\frac{1}{6}$ **(b) gradients multiply to −1, so perpendicular**
6.9	**(a)** $-1 \div \frac{1}{3} = -3$ **(b)** $m_1 = \frac{1}{2},\ m_2 = -2$ $\frac{1}{2} \times (-2) = -1$	**(a)** -3 **(b) gradients multiply to −1, so perpendicular**

6.10	**(a)** $-1 \div 3 = -\frac{1}{3}$ **(b)** $m_1 = 6,\ m_2 = -\frac{1}{6}$ $6 \times (-\frac{1}{6}) = -1$	**(a)** $-\frac{1}{3}$ **(b) gradients multiply to −1, so perpendicular**
6.11	**(a)** $-1 \div \frac{1}{4} = -4$ **(b)** $m_1 = -5,\ m_2 = \frac{1}{5}$ $-5 \times \frac{1}{5} = -1$	**(a)** -4 **(b) gradients multiply to −1, so perpendicular**
6.12	**(a)** $-1 \div (-2) = \frac{1}{2}$ **(b)** $m_1 = \frac{2}{3},\ m_2 = -\frac{3}{2}$ $\frac{2}{3} \times (-\frac{3}{2}) = -1$	**(a)** $\frac{1}{2}$ **(b) gradients multiply to −1, so perpendicular**

Number 3 · Using index laws (Lesson 6.1)

Q	Working	Answer
working out with powers		
7.1	**(a)** 2^{4+3} **(b)** $5^{3-1} = 5^2$ $= 25$ **(c)** $9^0 = 1$	**(a)** 2^7 **(b) 25** **(c) 1**
7.2	**(a)** 4^{2+5} **(b)** $3^{4-1} = 3^3$ $= 27$ **(c)** $12^0 = 1$	**(a)** 4^7 **(b) 27** **(c) 1**
7.3	**(a)** 6^{3+2} **(b)** $2^{5-1} = 2^4$ $= 16$ **(c)** $50^0 = 1$	**(a)** 6^5 **(b) 16** **(c) 1**

7.4	**(a)** 7^{5+2} **(b)** $10^{4-1} = 10^3$ $= 1000$ **(c)** $3^0 = 1$	**(a)** 7^7 **(b) 1000** **(c) 1**
7.5	**(a)** 5^{2+6} **(b)** $6^{3-1} = 6^2$ $= 36$ **(c)** $8^0 = 1$	**(a)** 5^8 **(b) 36** **(c) 1**
7.6	**(a)** 3^{5+4} **(b)** $2^{6-1} = 2^5$ $= 32$ **(c)** $11^0 = 1$	**(a)** 3^9 **(b) 32** **(c) 1**